S1 Supplementary Methods

Measurement contract and paired EEG--MEG feature analysis

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Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

S1.0 Dataset and measurement contract

[This report uses the healthy simultaneous-MEEG FacePerception H5 release `neuralmanifolddynamics_ds003645_20260722_215042`. Participant H5 files are parsed by subject and run, with empty-room files excluded. The analysis reads the H5 measurement contract and does not redo sensor preprocessing.]

S1.1 Epoch structure and feature export

[Each H5 exports `features_raw`, `features_robust_z`, and `features_projection_z`; the last is used for paired features because the H5 metadata documents degenerate robust-z MEG power slots. Exports also include `mnps_3d`, `coords_9d`, Jacobian layers, source flags, and twelve regional MNPS groups.]

S1.2 Primary metric definitions

[For a matched EEG/MEG feature family, temporal agreement is Pearson or Spearman correlation over valid concurrent time points. Run correlations are Fisher-transformed and averaged within participant. Mean-level difference is EEG mean minus MEG mean; its paired standardized effect is d_z.]

$$r_i = \mathrm{corr}\left(EEG_{i(t)}, MEG_{i(t)}\right) \tag{1}$$

S1.3 Statistical design

[Participant is the inference unit. Run means are averaged within participant. We use two-sided Wilcoxon signed-rank tests and 5,000 sign-flip permutations, with Benjamini–Hochberg FDR correction separately over global and regional feature families.]

S1.4 Regional geometry safeguard

[The six EEG regional MNPS arrays are numerically equivalent within each run, so they cannot estimate regional heterogeneity. Regional paired inference instead uses varying `__g_eeg_*` and `__g_meg_*` feature columns. The reachability wrapper reports MEG regional reachability only and records each skipped collapsed EEG run.]

Bibliography